**Hot Worked Low Carbon Co-28Cr-6Mo Alloy**

Hot worked R31537 cobalt is R31537 cobalt in the hot worked condition. The graph bars on the material properties cards below compare hot worked R31537 cobalt to other cobalt alloys (top) and the entire database (bottom). A full bar means this is the highest value in the relevant set. A half-full bar means it's 50% of the highest, and so on.

**Mechanical Properties**

Elastic (Young's, Tensile) Modulus

220 GPa32 x $10^6$ psi

Elongation at Break

14 %

Fatigue Strength

400 MPa58 x $10^3$ psi

Poisson's Ratio

0.29

Reduction in Area

13 %

Rockwell C Hardness

28

Shear Modulus

87 GPa13 x $10^6$ psi

Tensile Strength: Ultimate (UTS)

1140 MPa170 x $10^3$ psi

Tensile Strength: Yield (Proof)

780 MPa110 x $10^3$ psi

**Thermal Properties**

Latent Heat of Fusion

320 J/g

Melting Completion (Liquidus)

1360 °C2480 °F

Melting Onset (Solidus)

1290 °C2360 °F

Specific Heat Capacity

450 J/kg-K0.11 BTU/lb-°F

Thermal Conductivity

13 W/m-K7.3 BTU/h-ft-°F

Thermal Expansion

13 µm/m-K

**Electrical Properties**

Electrical Conductivity: Equal Volume

2.0 % IACS

Electrical Conductivity: Equal Weight (Specific)

2.1 % IACS

**Otherwise Unclassified Properties**

Density

8.4 g/cm$^3$530 lb/ft$^3$

Embodied Carbon

8.1 kg $CO_2$/kg material

Embodied Energy

110 MJ/kg46 x 10$^3$ BTU/lb

Embodied Water

530 L/kg63 gal/lb

**Common Calculations**

Resilience: Ultimate (Unit Rupture Work)

140 MJ/m$^3$

Resilience: Unit (Modulus of Resilience)

1380 kJ/m$^3$

Stiffness to Weight: Axial

15 points

Stiffness to Weight: Bending

24 points

Strength to Weight: Axial

38 points

Strength to Weight: Bending

29 points

Thermal Diffusivity

3.4 $mm^2/s$

Thermal Shock Resistance

28 points